Block Diagonal Representation Learning for Hyperspectral Band Selection

Shuying Li, *Member, IEEE*, Zhe Liu, Long Fang, and Qiang Li, *Member, IEEE*

***Abstract*— Hyperspectral band selection is viewed as an effective dimension reduction method. Recently, researchers present graph-based clustering for hyperspectral image (HSI) processing. However, most of them conduct clustering on a fixed data matrix so that it is sensitive to the quality of initial matrix. Moreover, these algorithms apply spectral clustering to obtain the final clustering result in increasing the time consumption. Based on these facts, we propose a block diagonal representation learning algorithm (BDRLA) in this article. BDRLA generates a high-quality similarity matrix by approximating the initial affinity matrix. Meanwhile, motivated to the spectral bands distance similarity matrix has a clear diagonal structure, a block diagonal similarity matrix with ordered partition points based upon the ℓ_2-norm is constructed. By doing so, the obtained similarity matrix is directly applied to subsequent processing without extracting the clustering indicators. Additionally, to estimate the importance of bands, dictionary learning (DL) is adopted to select the representative band in each cluster. Extensive experiment results on three public datasets indicate that the bands selected by the proposed method achieve satisfactory performance.**

***Index Terms*— Band selection, dictionary learning (DL), graph clustering, hyperspectral image (HSI), similarity matrix.**

NOMENCLATURE

D	3-D Hyperspectral image cube.
M, N, L	Height, width, and the number of bands of hyperspectral image.
$\boldsymbol{X}$	Hyperspectral data matrix.
$\boldsymbol{A}$	Affinity matrix.
m	Number of neighbors.
$\boldsymbol{a}_i^T$	ith row of the matrix $\boldsymbol{A}$.
$\boldsymbol{a}_{ij}$	ijth element of the matrix $\boldsymbol{A}$.
$\boldsymbol{S}$	Block diagonal similarity matrix.
K	Number of clusters.
$\boldsymbol{L}_S$	Laplacian matrix of the matrix $\boldsymbol{S}$.
$\boldsymbol{I}$	Identity matrix.
$\boldsymbol{1}$	All-one column vector.
$\boldsymbol{C}$	Coefficient matrix.
$\boldsymbol{S}_i$	ith subblock matrix.
$\|\|\cdot\|\|_{1,2}$	$\ell_{1,2}$-norm.
$Tr(\cdot)$	Trace of a matrix.
$\sigma_i(\cdot)$	ith smallest eigenvalues of a matrix.

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Shuying Li is with the School of Automation, Xi'an University of Posts and Telecommunications, Xi'an 710121, China, and also with the Shanghai Artificial Intelligence Laboratory, Shanghai 200232, China (e-mail: lishuying@xupt.edu.cn).

Zhe Liu is with the School of Automation, Xi'an University of Posts and Telecommunications, Xi'an 710121, China (e-mail: liuzhe@stu.xupt.edu.cn).

Long Fang is with the Northwest Institute of Nuclear Technology, Xi'an 710024, China (e-mail: fanglong@nint.ac.cn).

Qiang Li is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: liqmges@gmail.com).

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I. INTRODUCTION

HYPERSPECTRAL remote sensing image processing is a prominent technology that is developing in a variety of fields, including precision agriculture [1], mineral exploration [2], ecological environment monitoring [3], etc. Hyperspectral image (HSI) is obtained by the imaging spectrometer, which acquires information about objects of interest through large and narrow electromagnetic bands [4]. However, the high dimensionality of spectral bands and strong interband correlations bring about spectral information redundancy and high computational burden [5]. Moreover, the "Hughes" phenomenon occurs when an improbably large number of training samples are required to achieve high classification accuracy, which is also called "curse of dimensionality." The above dilemmas are conquered by dimension reduction, which is conducive to subsequent image analysis [6], [7].

Band selection [8] is regarded as an effective dimension reduction method. The essence of band selection is to select the distinctive and representative bands as the candidate bands and effectively improve the computational efficiency [9]. According to whether the training labels are available or not, band selection algorithms can be further classified into supervised [10], [11] and unsupervised [12], [13] methods. Specifically, supervised band selection algorithms select a band subset based on the class separability measure [14] of training samples and their class label. The significance of a band obtained by unsupervised algorithms is evaluated by various statistical measures [15]. Due to different training samples may exhibit different characteristics, it leads to the instability of the bands chosen by the supervised algorithms, which is undesirable for hyperspectral data analysis. Therefore, unsupervised band selection (UBS) is preferable for hyperspectral band selection.

The existing methods for UBS fall into two categories: ranking-based methods [16], [17], [18] and clustering-based methods [19], [20], [21]. The former evaluates the importance of each band according to a suitable criterion, such as

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information entropy [22], [23], mutual information [16], [24], local density [22], [25], etc. Then, the top-ranked bands in the sorted sequence are selected to form the subset. In contrast, the clustering-based methods are performed in the similarity space defined on the bands. Through partitioning the bands into disjoint clusters such that the bands in the same subset are more similar to each other than those in different clusters. These methods all treat each band as a data point and build a similarity matrix by Euclidean distance [26] or other criteria, and the bands close to the center of each cluster are picked to form the final band subset.

Among these band selection methods, the graph-based clustering methods appeal to our attention, e.g., spectral clustering [27], [28], ratio cut [29], normalized cut [30], low rank representation (LLR) [31], [32], and sparse subspace clustering (SSC) [33], [34]. The principle of graph clustering is how to construct a desirable high-quality similarity matrix (i.e., graph matrix). For instance, Sun et al. [35] utilize the angular similarity measurement to construct the similarity matrix in SSC model. Zhang et al. [36] propose a novel spectral–spatial SSC algorithm, which simultaneously considers the spectral and spatial properties of HSI to obtain a more accurate similarity matrix. However, these methods neglect the influence of image noise. Based on the above fact, Sun et al. [37] adopt correntropy measurement to construct the similarity matrix, and impose the sparsity and block diagonal constraint to improve the spectral bands clustering performance. Inspired by the above method, Sun et al. [38] also present a fast and latent low-rank subspace clustering method to improve the robustness of noise and reduce the algorithm computational complexity. It employs a correntropy metric to measure the similarity between pairwise bands and construct the similarity matrix. Zhang et al. [39] consider the difference between diverse homogeneous regions, and extract the spatial information of homogeneous regions in a structural graph by super-pixel segmentation. Unfortunately, these algorithms conduct clustering on a fixed input graph matrix so that it is sensitive to the quality of the initial graph matrix. Additionally, when performing graph-clustering algorithm, the hyperspectral bands are typically regarded as unordered and require post-processing to extract the clustering indicators in increasing the computational complexity.

Motivated by the above descriptions, it can be concluded that the hyperspectral bands should be viewed as ordered to deal with. Therefore, a block diagonal similarity matrix with exactly K clusters and data points have proper arrangement is learned, which is optimized by initial affinity matrix. Importantly, when selecting the representative bands in each cluster, the information content of the band is taken into account. To tackle this issue, dictionary learning (DL) is applied to select the bands by learning the optimal linear representation between the bands in each subblock matrix. In addition, as shown in Fig. 1, the similarity matrix acquired by calculating the Euclidean distance between the bands has a clear diagonal structure, which indicates that our algorithm is feasible further. Finally, a block diagonal representation learning algorithm (BDRLA) for hyperspectral band selection

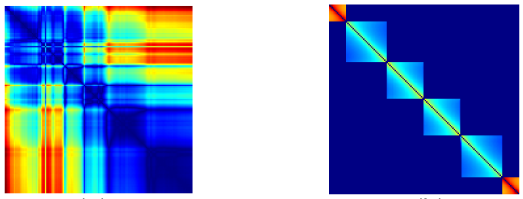


Fig. 1. Difference between distance similarity matrix and block diagonal similarity matrix on Botswana dataset. (a) Euclidean distance similarity matrix. (b) Block diagonal similarity matrix.

BDRLA is proposed. The contributions of this article are summarized as follows.

1) BDRLA considers the correlation between the adjacent bands to construct the initial graph model of all bands. Moreover, based on the fact that Euclidean distance similarity matrix has a diagonal structure, the block diagonal similarity matrix is constructed by graph clustering.
2) To obtain a high-quality similarity matrix, BDRLA allows the initial graph matrix itself to be adjusted adaptively when performing clustering. Meanwhile, the learned similarity matrix possesses the strict block diagonal structure with K clusters in greatly reducing computational cost.
3) An optimal linear combination method between the spectral bands is designed by DL to estimate the importance of bands in each subblock matrix, which can achieve the low-dimensional manifold representation of the band vectors.

The remainder of this article is organized as follows. Section II briefly reviews the rationale of graph clustering and DL. In Section III, the detailed descriptions of the proposed method are presented. The experiments and results on three real hyperspectral datasets are shown in Section IV. Finally, conclusion is given in Section V.

II. Related Work

In this section, a brief discussion about relevant techniques for the proposed algorithm, including graph clustering and DL.

A. Graph Clustering

Graph-based clustering [40], [41] is thoroughly studied and many different algorithms are investigated. It aims to group vertices according to their characteristics such that there is high intracluster similarity and low intercluster similarity [42]. Formally, a clustering algorithm divides a graph $G = (V, E)$ into k clusters $\psi = C_1, C_2, \ldots, \ldots, C_k$ in which a vertex $\boldsymbol{v} \in \psi_i$ are closer to all vertices in cluster C_i than the vertices in other clusters. Here, V and E denote the set of vertices and edges, respectively. Next, three traditional graph-based clustering algorithms are introduced in detail, including spectral clustering, SSC, and LLR.

1) Spectral Clustering: Given a data matrix $\boldsymbol{B} = [\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_n] \in \mathbb{R}^{d\times n}$, in which d and n are the dimension of one sample and the number of data points, respectively. Conventional spectral clustering first computes an $n \times n$ affinity matrix, in which each entry corresponds to the similarity

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of two objects according to some similarity metrics. Then, the eigen decomposition is performed on the graph Laplacian matrix of the affinity matrix to obtain the k eigenvectors associated with the first k eigenvalues. Finally, embedding the data into the low-dimensional space via the obtained k eigenvectors, and the final clustering can be achieved via k-means or other discretization techniques.

2) Sparse Subspace Clustering: Let $\boldsymbol{Y} = [\boldsymbol{y}_1, \boldsymbol{y}_2, \ldots, \boldsymbol{y}_n] \in \mathbb{R}^{d\times n}$ denote a data matrix, the corresponding sparse self-representation matrix $\boldsymbol{Z} \in \mathbb{R}^{n\times n}$ could be solved by the following minimization problem:

$$\min_{Z} \frac{1}{2}||\boldsymbol{Y} - \boldsymbol{Y}\boldsymbol{Z}||_2^2 + \mu||\boldsymbol{Z}||_1 \quad \text{s.t. } \text{diag}(\boldsymbol{Z}) = 0 \tag{1}$$

where $\mu > 0$ is a tradeoff parameter and $||\cdot||_{1,2}$ represents the $\ell_{1,2}$-norm. The constraint $\text{diag}(\boldsymbol{Z}) = 0$ is the vector diagonal entries of matrix $\boldsymbol{Z}$, which is used to eliminate the trivial solution that each data point is simply a linear combination of itself. The ideal solution of (1) finds a set of data points from a single subspace. It guarantees that the coefficient matrix $\boldsymbol{Z}$ is sparse or block diagonal and could benefit the correct segmentation of all data points into separate subspaces.

Then, the matrix $\boldsymbol{Z}$ is utilized to construct the similarity matrix $\boldsymbol{W}$ for inferring the segmentation of all data points into different subspaces. The matrix $\boldsymbol{W}$ can be regarded as an undirected weighted graph, where each entry $W_{i,j}$ represents the weight of the edges between pairwise points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. In the end, the SSC methods utilize the l_1-directed graph construction (DGC) measurement to guarantee the symmetrization of weights between pairwise points $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$. The DGC measurement is performed as follows:

$$W_{i,j} = \frac{|Z_{i,j}| + |Z_{j,i}|}{2} \tag{2}$$

where $|Z_{i,j}|$ is the absolute value of the entry $Z_{i,j}$ in matrix $\boldsymbol{Z}$. After constructing the similarity matrix $\boldsymbol{W}$, the SSC uses spectral clustering to cluster all data points into their underlying subspaces.

3) Low Rank Representation: LLR algorithms aim at seeking the lowest rank representation of all data points concurrently, such that each of data point can be represented as a linear combination of certain bases. The low-rank regularizer in LRR model has a strong connection to the recent theoretical advances on robust principal component analysis [43], which leads to novel and powerful modeling options for many applications. It has been demonstrated that the LRR model can efficiently capture the underlying low-dimensional structures of dataset.

Given a data matrix $\boldsymbol{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \ldots, \boldsymbol{h}_n] \in \mathbb{R}^{d\times n}$ be a collection of n data points, the LRR model is formulated as a following minimization problem:

$$\min \text{rank}(\boldsymbol{Q}) \quad \text{s.t. } \boldsymbol{H} = \boldsymbol{H}\boldsymbol{Q} \tag{3}$$

where $\boldsymbol{Q}$ denotes the lowest rank representation matrix under a self-expressive dictionary to SSC. However, this minimization problem is quite challenging to solve in general, thus the nuclear norm $||\boldsymbol{Q}||_*$ is adopted to be a surrogate for the rank of matrix $\boldsymbol{Q}$. Under the reconstruction error of Euclidean metric, the final LRR model is presented by an unconstrained optimization problem as follows:

$$\min ||\boldsymbol{Q}||_* + \omega||\boldsymbol{H} - \boldsymbol{H}\boldsymbol{Q}||_F^2 \tag{4}$$

where ω denotes a penalty parameter to balance the impact of these two terms, and $||\cdot||_*$ and $||\cdot||_F$ represent the Nuclear norm and Frobenius norm, respectively.

LRR methods can discover the low-dimensional structure of data that indicates the similarity between data points. Unlike the SSC model, which utilizes sparsity constraint while reconstructing each data point independently, the LRR model reconstructs the whole data in a more comprehensive manner.

Graph learning [44] can effectively express the essential structure and even the complex higher order relationships of data points, which demonstrates impressive performance and great potential due to the flexible construction of a similarity graph. Generally speaking, the graph-based clustering methods represent the collection of objects as a graph in which the (dis)similarity relationships among different objects are measured by the similarity graph. These methods typically employ a two-step strategy. First, a similarity graph (i.e., a similarity matrix) is constructed in which the data points are the nodes and the relationships are the weights, and it can be computed using various similarity criteria. Then, the clustering can be achieved based on the graph theory optimization process. It should be emphasized that the quality of the similarity graph plays a decisive role in the final clustering performance.

B. Dictionary Learning

DL [15], [45] is a prominent mechanism to describe images, which practices in numerous image-processing applications like image super-resolution [46], image denoising, data compression, etc. The formal title of DL and sparse representation (SR) in academic circles should be Sparse DL. DL is a kind of SR of the data matrix obtained on a dictionary basis space. Based on the SR framework, the image data points can be sparsely and linearly represented by some dictionary bases commonly. It mainly consists of two stages: dictionary generation and sparse coding with precomputed dictionary.

As per the notion of SR theory [47], a sparse image can be represented as a linear combination of the smallest possible number of columns of an overcomplete dictionary. An arbitrary image patch extracted from a given image is modeled using a column vector. A dictionary composes of column vectors and is represented by $\boldsymbol{D} \in \mathbb{R}^{n\times H}$, if $n < H$ the dictionary is an overcomplete entity. SR theory presumes $x = D\alpha$, with $\alpha \in \mathbb{R}^H$ is a sparse coefficient vector of $\boldsymbol{x}$ with the associated dictionary $\boldsymbol{D}$. Since the dictionary is redundant, there exist several $\boldsymbol{x}$-vectors that satisfy $\boldsymbol{x} = D\alpha$. SR aims to find the sparsest vector that can effectively represent an image patch ($\boldsymbol{x}$). The SR model can be achieved by solving the following optimization problem:

$$\hat{\alpha} = \arg\min_{\alpha} ||\alpha||_1 + \phi\varphi(\alpha) \quad \text{s.t. } ||x - \boldsymbol{D}\alpha||_2^2 \le \varepsilon. \tag{5}$$

Here, ϕ denotes the error tolerance in the form of upper bound for sparse approximation error $||x - \boldsymbol{D}\alpha||_2^2$, and $\varphi(\alpha)$ stands for a sparsity-inducing function including the generalized

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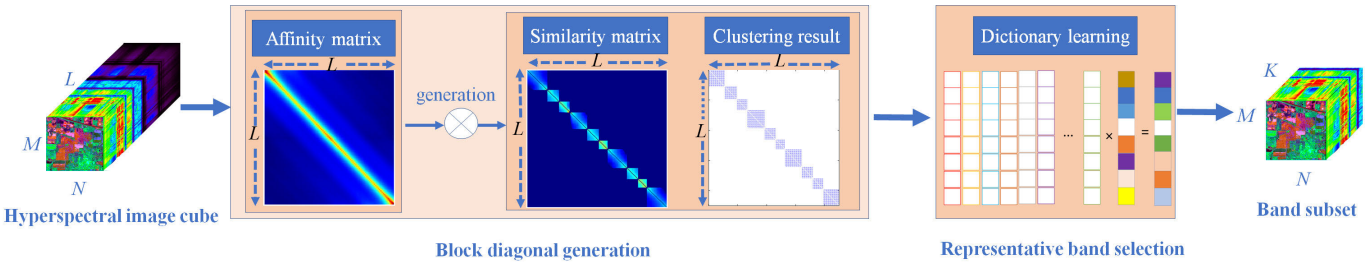


Fig. 2. Overall workflow of the proposed method. All HSI data points (i.e., bands) are partitioned into K clusters orderly by constructing the block diagonal similarity matrix. Then, DL is adopted to select the representative band in each subblock matrix.

minimax-concave penalty and ℓ_1-norm. This minimization problem can be resolved by an iterative algorithm efficiently, such as orthogonal matching pursuit (OMP) and alternating direction method of multipliers (ADMM).

III. Methodology

This section introduces the proposed algorithm for hyperspectral band selection elaborately, namely BDRLA, whose workflow is depicted in Fig. 2. The fundamental purpose of the proposed method is to utilize HSI data points (i.e., spectral bands) to construct a block diagonal similarity matrix composed of K subblock matrices (i.e., K clusters). Then, DL is employed to select the most representative band in each subblock matrix. For easy reference and to enhance the readability of the article, the major notations used in this article are summarized in Nomenclature.

A. Diagonal Block Generation via Rank Constraint

1) Learning an Affinity Matrix: Let $D \in \mathbb{R}^{M\times N\times L}$ denote the HSI cube, where L is the number of total spectral band images, and the width and height of each band image are M and N, respectively. The matrix of each spatial image band is stretched to a 1-D vector. Thus, we get the matrix $\boldsymbol{X} = [\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_L] \in \mathbb{R}^{MN\times L}$, where $\{\boldsymbol{x}_i\}_{i=1}^{L}$ is the vector of the ith band image. The affinity value of matrix $\boldsymbol{A}$ should satisfy such that smaller distance between data points corresponds to a larger affinity value $\boldsymbol{a}_{ij}$. In addition, to get a more accurate affinity matrix $\boldsymbol{A}$, the constraint $\boldsymbol{a}_{ii} = 0$ should be added to avoid the trivial solution. Meanwhile, to avoid the case that some rows are all zero, we further introduce an essential constraint $\boldsymbol{a}_i^T\mathbf{1} = 1$ such that each row of the matrix $\boldsymbol{A}$ sums to one. The final objective function is proposed as

$$\min_{\boldsymbol{a}_i^T\mathbf{1}=1,\boldsymbol{a}_i\geq 0} \sum_{j=1}^{L} ||\boldsymbol{x}_i - \boldsymbol{x}_j||_2^2 \boldsymbol{a}_{ij} + \gamma \sum_{j=1}^{L} \boldsymbol{a}_{ij}^2 \tag{6}$$

where γ is the parameter used for adjusting the weight of regularization term.

In most cases, matrix $\boldsymbol{A}$ is as sparse as possible for efficiency and higher performance. Therefore, affinity value should satisfy the maximal γ such that the optimal solution $\boldsymbol{a}_i$ has exactly m nonzero values, i.e., $||\hat{\boldsymbol{a}}_i||_0 = m$. By denoting $\boldsymbol{d}_{ij} = ||\boldsymbol{x}_i - \boldsymbol{x}_j||_2^2$ and $\boldsymbol{d}_i$ as a vector with the jth element as $\boldsymbol{d}_{ij}$, (6) could be simplified as

$$\min_{\boldsymbol{a}_i^T\mathbf{1}=1,\boldsymbol{a}_i\geq 0} \frac{1}{2}\left\|\boldsymbol{a}_i + \frac{\boldsymbol{d}_i}{2\gamma}\right\|_2^2. \tag{7}$$

The Lagrange algorithm is used to solve the above equation as follows:

$$\mathcal{L}(\boldsymbol{a}_i, \eta, \beta_i) = \frac{1}{2}\left\|\boldsymbol{a}_i + \frac{\boldsymbol{d}_i}{2\gamma}\right\|_2^2 - \eta\left(\boldsymbol{a}_i^T\mathbf{1} - 1\right) - \beta_i^T\boldsymbol{a}_i \tag{8}$$

where η and β_i are the Lagrange multipliers. The optimal solution $\hat{\boldsymbol{a}}$ should satisfy that the derivative of (8) is zero. Therefore, for the jth element of $\hat{\boldsymbol{a}}_i$, the equality is computed as follows:

$$\hat{\boldsymbol{a}}_{ij} + \frac{\boldsymbol{d}_{ij}}{2\gamma} - \eta - \beta_{ij} = 0. \tag{9}$$

After obtaining the $a_{ij}\beta_{ij} = 0$ according to the KKT condition, (9) can be transformed into

$$\hat{\boldsymbol{a}}_{ij} = \left(-\frac{\boldsymbol{d}_{ij}}{2\gamma} + \eta\right)_+. \tag{10}$$

Based on the constraint $||\hat{\boldsymbol{a}}_i||_0 = m$, we get $\hat{\boldsymbol{a}}_{im} > 0$ and $\hat{\boldsymbol{a}}_{i,m+1} = 0$. Similarly, according to the constraint $\boldsymbol{a}_i^T\mathbf{1} = 1$, the following equality for η is acquired:

$$\sum_{j=1}^{m}\left(-\frac{\boldsymbol{d}_{ij}}{2\gamma} + \eta\right) = 1 \Rightarrow \eta = \frac{1}{m} + \frac{1}{2m\gamma}\sum_{j=1}^{m}\boldsymbol{d}_{ij}. \tag{11}$$

Hence, the inequality for γ is obtained according to (10) and (11) as follows:

$$\frac{m}{2}\boldsymbol{d}_{im} - \frac{1}{2}\sum_{j=1}^{m}\boldsymbol{d}_{ij} < \gamma < \frac{m}{2}\boldsymbol{d}_{i,m+1} - \frac{1}{2}\sum_{j=1}^{m}\boldsymbol{d}_{ij}. \tag{12}$$

To obtain the optimal solution $\hat{\boldsymbol{a}}_i$ that has exactly m nonzero values, the γ is defined as

$$\gamma = \frac{m}{2}\boldsymbol{d}_{i,m+1} - \frac{1}{2}\sum_{j=1}^{m}\boldsymbol{d}_{ij}. \tag{13}$$

Finally, the optimal affinity value $\hat{\boldsymbol{a}}_{ij}$ can be denoted as follows:

$$\hat{\boldsymbol{a}}_{ij} = \begin{cases} \dfrac{\boldsymbol{d}_{i,m+1} - \boldsymbol{d}_{ij}}{m\boldsymbol{d}_{i,m+1} - \sum_h^m \boldsymbol{d}_{ih}}, & j \leq m \\ 0, & j > m. \end{cases} \tag{14}$$

The affinity value $\hat{\boldsymbol{a}}_{ij}$ has the following advantages: 1) this optimization process only involves the basic operations of

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addition, subtraction, multiplication, and division, so the computational efficiency is lower than the current method; and 2) the affinity values are distance consistent. If the distance between $\boldsymbol{x}_i$ and $\boldsymbol{x}_j$ is smaller than the distance between $\boldsymbol{x}_i$ and $\boldsymbol{x}_l$, then the affinity value $\hat{\boldsymbol{a}}_{ij}$ is larger than the affinity value $\hat{\boldsymbol{a}}_{il}$. That facilitates the construction of a better similarity matrix in Section III-A2, improving thereby the performance of the graph clustering.

2) Construction of Similarity Matrix: With the above affinity matrix $\boldsymbol{A}$, we learn a similarity matrix $\boldsymbol{S}$ with several subblock matrices via low rank constraint (i.e., $\text{rank}(\boldsymbol{L}_S) = L - K$), where $\boldsymbol{L}_S$ and K are the corresponding Laplacian matrix and the number of connected components in matrix $\boldsymbol{S}$, respectively. Equally, to avoid the situation that some rows of matrix $\boldsymbol{S}$ are all zeros, we further constrain the matrix $\boldsymbol{S}$ such that the sum of each row is one, i.e., $\sum_j \boldsymbol{s}_{ij} = 1$.

Under these constraints, the learned matrix $\boldsymbol{S}$ could best approximate the initial affinity matrix $\boldsymbol{A}$. This minimization problem can be formulated into the model in

$$\min_{\sum_j \boldsymbol{s}_{ij}=1, rank(\boldsymbol{L}_S)=L-K} ||\boldsymbol{S}-\boldsymbol{A}||_2^2 + 2\lambda \sum_{i=1}^{K} \sigma_i(\boldsymbol{L}_S) \tag{15}$$

where λ is a balancing parameter. The constraint term $\sum_{i=1}^{K} \sigma_i(\boldsymbol{L}_S)$ is the block diagonal regularization term whose $\sigma_i(\boldsymbol{L}_S)$ denotes the ith smallest eigenvalues in matrix $\boldsymbol{L}_S$. Since the constraint term $\text{rank}(\boldsymbol{L}_S) = L - K$ is a complex nonlinear constraint and matrix $\boldsymbol{L}_S$ depends on matrix $\boldsymbol{S}$, it is extremely difficult to solve the aforementioned problem. We propose an efficient optimization algorithm to solve the minimization problem (15) in the next section.

3) Algorithm Optimization: An interesting proposition is proven in Luxburg's work [48], that is, the multiplicity of the eigenvalue zero of the corresponding Laplacian matrix $\boldsymbol{L}_S$ equals the number of connected components in matrix $\boldsymbol{S}$. According to the above description, the corresponding equality is defined as

$$\sum_{i=1}^{K} \sigma_i(\boldsymbol{L}_S) = \min_{\boldsymbol{F}\in\mathbb{R}^{L\times K}, \boldsymbol{F}^T\boldsymbol{F}=\boldsymbol{I}} Tr\left(\boldsymbol{F}^T \boldsymbol{L}_S \boldsymbol{F}\right) \tag{16}$$

where $Tr(\cdot)$ represents the trace of matrix, and matrix $\boldsymbol{F}$ is composed of K eigenvectors corresponding to the Laplacian matrix $\boldsymbol{L}_S$.

When λ is a constant number, noting that $\sigma_i(\boldsymbol{L}_S) \geq 0$ for every i, thus the optimal solution $\boldsymbol{S}$ makes the second term equal to zero and the constraint $rank(\boldsymbol{L}_S) = L - K$ is satisfied. Finally, our proposed model is described by solving the following optimization problem:

$$\min_{\sum_j \boldsymbol{s}_{ij}=1, \boldsymbol{s}_{ij}\geq 0} ||\boldsymbol{S}-\boldsymbol{A}||_2^2 + 2\lambda Tr\left(\boldsymbol{F}^T \boldsymbol{L}_S \boldsymbol{F}\right). \tag{17}$$

Compared with the original equation, (17) is much easier to solve. For each i, the minimization function in (17) can be transformed as

$$\min_{\sum_j \boldsymbol{s}_{ij}=1, \boldsymbol{s}_{ij}\geq 0} \sum_j \left(\boldsymbol{s}_{ij} - \boldsymbol{a}_{ij}\right)^2 + \lambda \sum_j ||\boldsymbol{f}_i - \boldsymbol{f}_j||_2^2 \boldsymbol{s}_{ij}. \tag{18}$$

To simplify the process, we denote $\boldsymbol{z}_{ij} = ||\boldsymbol{f}_i - \boldsymbol{f}_j||_2^2$ and $\boldsymbol{z}_i$ as a vector with the jth element equal to $\boldsymbol{z}_{ij}$ (similarly for $\boldsymbol{s}_i$ and $\boldsymbol{a}_i$), (18) can be formulated in vector form as

$$\min_{\boldsymbol{s}_i^T \boldsymbol{1}=1, \boldsymbol{s}_i\geq 0} \left\| \boldsymbol{s}_i - \left(\boldsymbol{a}_i - \frac{\lambda}{2}\boldsymbol{z}_i\right)\right\|_2^2. \tag{19}$$

This equation can be solved with a closed form solution as in (7).

B. Representative Band Selection via DL

When performing the representative band selection, the traditional methods usually select the most relevant or informative band in each cluster. However, this strategy made for the selected bands is discriminative. To handle this problem, an optimal linear combination between the band vectors is obtained by DL for each submatrix to select the representative bands.

With respect to the block diagonal similarity matrix $\boldsymbol{S} = [\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_K]$, each subblock matrix $\{\boldsymbol{S}_i\}_i^K \in \mathbb{R}^{L_i \times L_i}$ is often treated independently, where L_i represents the size of each subblock matrix. This placement of L data points into the K subblock matrices also implies $\sum_i^K L_i = L$. Accordingly, each data point $\boldsymbol{s}_i \in \mathbb{R}^{L_i \times 1}$ belonging to $\boldsymbol{S}_i$ can be represented as $\boldsymbol{s}_i = \boldsymbol{S}_i \boldsymbol{c}_i$, where $\boldsymbol{c}_i \in \mathbb{R}^{L_i \times 1}$ and $\boldsymbol{S}_i$ are the coefficient vector and dictionary of the data points $\boldsymbol{s}_i$, respectively. Based on supposing the subblock matrix $\boldsymbol{S}_i$ is the dictionary matrix while processing the DL, this problem is solved by the following objective function:

$$\min_{\boldsymbol{c}_i \geq 0} ||\boldsymbol{s}_i - \boldsymbol{S}_i \boldsymbol{c}_i||_2^2 + \delta ||\boldsymbol{c}_i||_1 \tag{20}$$

where δ is the sparsity regularization constant. After obtaining the sparse vector $\boldsymbol{c}_i$ of each band vector, the coefficient matrix $\{\boldsymbol{C}_i\}_i^K$ is generated simultaneously. Moreover, to select the most representative band in each cluster, the band corresponding to the maximum sum of each column in matrix $\boldsymbol{C}_i$ is selected to form the band subset. Finally, the whole pseudocode of the proposed band selection algorithm is outlined in Algorithm 1.

C. Computational Complexity Analysis

The implementation of the proposed method mainly consists of two steps: block diagonal generation and representative band selection. In the following, we analyze the time complexity of the two parts in the proposed method separately.

1) Block Diagonal Generation: The block diagonal generation includes two stages: learn an affinity matrix and construct a block diagonal similarity matrix. Specifically, the affinity matrix is learned by calculating the Euclidean distance between HSI data points. The time complexity of this stage costs $\mathcal{O}(L)$, where L represents the entire band vectors of hyperspectral data matrix. The latter mainly depends on updating the block diagonal similarity matrix, the computational complexity is approximately $\mathcal{O}(L + lL^2)$, where l denotes the number of iterations. Apparently, the time complexity of the latter is higher than the former.

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Algorithm 1 Framework of BDRLA

Input: HSI data matrix $\boldsymbol{X} = \{\boldsymbol{x}_i\}_i^L \in \mathbb{R}^{MN\times L}$, the number of neighbors m, and two constant numbers λ and δ.

Output: the band subset $\boldsymbol{M}(:, K)$.

step1: Learning an initial affinity matrix A by Eq. (6).
for $i = 1 \quad to \quad L$ **do**
  if $j \le m$ **then**
    $\hat{a}_{ij} = \frac{\boldsymbol{d}_{i,m+1} - \boldsymbol{d}_{ij}}{m\boldsymbol{d}_{i,m+1} - \sum_h^m \boldsymbol{d}_{ih}}$
  else
    $\hat{a}_{ij} = 0$
  end
end
step2: Construct the similarity matrix S by Eq. (15).
step3: Solve the BDRLA algorithm (15).
  Initialization $\boldsymbol{F} \in \mathbb{R}^{L\times K}$, $\boldsymbol{F}^T\boldsymbol{F} = \boldsymbol{I}$, $\lambda = 0.1$.
while *not converge* **do**
  1) Update $\boldsymbol{F}$ using the Eq. (17), which is formed by the K eigenvectors of matrix $\boldsymbol{L}_S$ corresponding to the K smallest eigenvalues;
  2) Update the i-th row of matrix $\boldsymbol{S}$ by solving the problem (19);
end
  The obtained similarity matrix $\boldsymbol{S} = [\boldsymbol{S}_1, \boldsymbol{S}_2, \ldots, \boldsymbol{S}_K]$.
step4: Representative band selection in each subblock.
  for $i = 1 \quad to \quad K$ **do**
  1) Update $\boldsymbol{c}_i$ by solving the optimization problem (20);
  2) Select the band whose corresponding the maximum sum of matrix $\boldsymbol{C}_i$ column vectors as the representative band.
end
return subset $\boldsymbol{M}$.

2) Representative Band Selection: To obtain the desired band in each subblock matrix (i.e., cluster), we utilize the DL to gain the linear representation between vectors in each subblock matrix, and select the band corresponding to the vector with the largest sum of columns. For each subblock matrix $\{\boldsymbol{S}_i\}_i^K \in \mathbb{R}^{L_i\times L_i}$, the time complexity of per DL is $\mathcal{O}(TL_i)$, where T is the number of iterations for per DL. The total complexity of K subblock matrices takes $\mathcal{O}(TL)$. Therefore, the overall complexity of the proposed method is $\mathcal{O}((2+T)L + lL^2)$.

IV. Experiment

In this section, we implement numerous experiments to assess the superiority of the proposed method. Three real-world HSI datasets are first introduced. Then, the experimental settings are described, including the classifier setting, comparison methods, the number of selected bands, and evaluation metrics. Finally, the effectiveness and advantages of the proposed method are verified from several aspects in detail.

A. Datasets

1) Pavia University: The Pavia University dataset in 2002 was collected by ROSIS sensor during flight campaign. This HSI has 115 bands with the size of 610 × 340 pixels. This dataset removes some bands with low signal-to-noise ratio and obtains 103 spectral bands with nine classes.

2) Botswana: The Botswana dataset in 2001 was acquired by the NASA EO-1 satellite over the Okavango Delta. It contains 242 spectral bands ranging from 400 to 2500 nm. After excluding some noisy bands, the remaining 145 spectral bands were employed for his classification.

3) Salinas: The Salinas dataset was gathered by AVIRIS in 1998. It contains 224 bands with 10-nm widths and wavelengths ranging from 400 to 2500 nm. For classification purposes, 16 classes of ground object were defined.

B. Experiment Setting

1) Classifier Setting: Two classifiers, k-nearest neighbor (KNN) and support vector machine (SVM), are adopted to classify samples on three public hyperspectral datasets to verify the quality of the obtained bands. On different datasets, the two classifiers have identical parameter settings in our experiments. For all classification validation studies, the number of neighbors in KNN is set to be 5. The radial basis function (RBF) kernel is used for the SVM classifier, and the penalty C and gamma are initialized to 1×10^4 and 0.5, respectively. Furthermore, since the above two classifiers are supervised, 10% samples of the selected bands are randomly picked as training set and the remaining samples are employed to implement the test task. It should be noted that the random selection of these samples may cause the classification results to be unstable. To eliminate the influence, the final classification results are obtained by calculating the average value after five runs.

2) Comparison Methods: Five UBS methods are compared with our method in multiple dimensions to demonstrate the superiority of the proposed method. Furthermore, for performance comparison, all bands on three hyperspectral dataset are taken into account. The comparison algorithms detail are as follows.

a) Maximum-Variance Principal Components Analysis (MVPCA) [18]: The covariance matrix of the bands used as an eigenvalue to construct a loading factors matrix. All bands are sorted according to the loading factors matrix.

b) E-FDPC [49]: After all bands are clustered by similarity matrix, the local density and intraclass distance of each band are computed and normalized to rank the bands.

c) Fast neighborhood grouping method for hyperspectral band selection (FNGBS) [50]: All bands are grouped by coa-e–fine division strategy. Then, the band with the largest product is gained by local density and information entropy in each group.

d) OPBS [51]: The bands with the maximum variance as the initial selected band set. Then calculating the orthogonal projections (OPs) of all the candidate bands and tries to find the band with the largest OP as the desired band.

e) VGBS [52]: After performing singular value decomposition (SVD) to reduce the dimensionality of band image, the

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TABLE I
CLASSIFICATION RESULTS BY SETTING DIFFERENT PARAMETERS m AND λ ON PAVIA UNIVERSITY DATASET USING SVM CLASSIFIER(%)

m	λ	5	7	10	12	15	20
10	0.05	80.73	84.07	87.82	**91.49**	90.60	92.35
	0.1	81.84	82.76	87.63	91.10	90.63	92.30
	0.2	82.10	81.80	87.78	90.31	90.62	**92.66**
	1	80.18	85.46	87.45	89.80	89.03	90.10
20	0.05	84.30	84.19	87.91	91.14	89.03	91.34
	0.1	84.33	84.77	87.99	91.16	89.13	91.71
	0.2	84.50	84.63	88.01	91.05	89.14	92.16
	1	**84.76**	**87.58**	**90.96**	90.88	**91.38**	91.80
30	0.05	82.71	85.28	88.70	91.13	89.81	90.58
	0.1	82.35	85.46	87.70	91.14	89.17	91.52
	0.2	81.31	85.37	88.74	91.04	89.33	91.11
	1	83.47	84.07	87.98	90.91	89.40	90.78

TABLE II
CLASSIFICATION RESULTS BY SETTING DIFFERENT PARAMETERS m AND λ ON BOTSWANA DATASET USING SVM CLASSIFIER(%)

m	λ	5	7	10	12	15	20
10	0.05	81.80	83.61	86.08	87.35	88.01	88.96
	0.1	81.80	83.51	84.64	88.58	87.76	88.62
	0.2	81.66	81.70	84.57	86.87	86.84	89.24
	1	75.22	82.76	85.09	89.00	88.31	88.28
20	0.05	79.19	83.79	86.94	88.52	87.04	87.42
	0.1	80.08	83.75	86.53	87.93	**89.24**	89.48
	0.2	79.64	84.88	86.60	87.66	88.55	88.89
	1	79.73	84.13	88.10	87.28	87.35	89.17
30	**0.05**	**82.55**	**84.29**	**87.45**	88.38	88.35	**90.06**
	0.1	81.38	84.27	86.32	**89.20**	88.04	88.34
	0.2	81.83	82.79	86.87	87.18	87.28	89.89
	1	76.20	83.00	86.18	88.38	87.62	89.54

band with the shortest hyperplane distance spanning the other bands is removed by sequential backward selection.

3) Number of Selected Bands: When performing hyperspectral band selection, how many bands to choose in practice is unknown. Typically, different number of bands are set to analyze accuracy through classifiers. In our experiment, we set the range of [5, 50] to investigate the impact of the different numbers of band.

4) Evaluation Metrics: To evaluate the performance of the proposed algorithm, two widely used methods of classification accuracy measurement are conducted for all datasets: overall accuracy (OA) and average OA (AOA).

C. Parameter Analysis

To investigate the performance of the proposed method, we analyze the influence of different parameters, including the parameters m, λ, and δ. The first two parameters are applied in block diagonal generation process, the parameter δ is used in representative band selection part. Therefore, this section is divided into two branches: parameter analysis of block diagonal generation and parameter analysis of representative band selection. Additionally, the optimal classification results of OA are bold in each table.

1) Parameter Analysis of Block Diagonal Generation: In the proposed band selection method, before constructing the block diagonal similarity matrix, the number of neighbors m is adopted for computing the affinity value. The parameter m is an integer and is easy to tune. Moreover, a constant number λ is fundamental to form the block diagonal similarity matrix. In this section, we set different parameters to study the impact on the performance of the proposed method when different number of bands are selected (the range of [5, 20]). Specifically, we take parameter λ to 0.05, 0.1, 0.2, and 1 for all datasets without further tuning. With respect to parameter m, it is manually set to 10, 20, and 30, respectively.

a) Pavia university: The classification results about different parameter values are depicted in Table I. It can be seen that in this table these parameters greatly change the classification results, particularly as for m. The reason may be that the final classification accuracy is largely affected by the initial affinity matrix. Experiments reveal the parameters ($m = 20$ and $\lambda = 1$) can obtain superior performance. Therefore, in our work, we empirically set the parameters m and λ to 20 and 1, respectively.

b) Botswana: Table II illustrates the performance obtained by varying different parameters on Botswana dataset by SVM classifier. According to the table, when the parameter

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TABLE III
CLASSIFICATION RESULTS BY SETTING DIFFERENT PARAMETERS m AND λ ON SALINAS DATASET USING SVM CLASSIFIER(%)

m	λ	5	7	10	12	15	20
10	0.05	88.73	90.22	90.66	90.75	91.35	91.83
	0.1	88.70	**90.27**	90.40	91.09	91.38	91.76
	0.2	88.66	90.17	90.76	90.87	91.21	91.76
	1	**89.46**	90.20	**90.89**	90.87	**91.51**	91.80
20	0.05	87.61	88.11	90.08	90.72	91.31	91.65
	0.1	87.65	87.83	90.12	90.61	91.35	91.55
	0.2	87.49	87.82	89.90	90.69	91.30	91.75
	1	87.61	88.10	90.03	**91.14**	91.11	**91.85**
30	0.05	89.00	89.51	90.55	90.60	91.51	91.64
	0.1	88.72	89.82	90.52	90.75	91.28	91.30
	0.2	88.74	89.38	90.04	90.66	91.53	91.84
	1	88.75	89.60	90.42	90.37	91.26	91.60

TABLE IV
CLASSIFICATION RESULTS BY SETTING DIFFERENT PARAMETERS δ ON THREE DATASETS USING SVM CLASSIFIER(%)

Dataset	δ	5	7	10	12	15	20
Pavia University (m=20, λ=1)	0.05	83.73	84.22	87.95	90.05	89.78	91.84
	0.1	79.15	84.85	88.16	**90.65**	89.81	92.05
	0.5	**84.74**	**85.35**	**88.37**	89.89	**90.32**	**92.58**
	1	81.84	85.27	88.15	90.13	89.80	92.36
Botswana (m=30, λ=0.05)	**0.05**	**83.10**	**85.46**	85.43	**89.41**	**88.93**	**90.12**
	0.1	82.21	83.65	85.53	88.76	87.25	90.06
	0.5	82.17	82.48	85.12	87.04	88.39	89.92
	1	82.72	81.80	**86.18**	86.90	87.25	90.05
Salinas (m=10, λ=1)	**0.05**	**89.61**	90.07	**90.77**	**90.95**	**91.37**	**91.97**
	0.1	84.24	**90.15**	90.38	90.81	91.06	91.62
	0.5	86.14	88.78	90.16	90.62	91.15	91.21
	1	87.82	89.89	89.96	90.69	91.37	91.54

λ is fixed, the parameter m has little influence on classification accuracy. On the contrary, it can be seen that setting different parameters λ displays considerable change, especially while selecting a few bands. The reason for this phenomenon may be that the image size is too large to perform effective block diagonal clustering. Overall, when the parameters m and λ are set to 30 and 0.05, respectively, the classification performance is relatively superior.

c) Salinas: Likewise, we also set the same parameters to analyze the impact on Salinas dataset. Table III provides the classification accuracy of selecting different number of bands. One can observe that regardless of how many the parameter m is set, there is little fluctuation in OA. Compared with the Botswana dataset, the classification accuracy is not very different by setting several parameters λ. It can be concluded that this dataset is less sensitive to these two parameters. Thus, the parameters m and λ are determined to be 10 and 1 in our experiment, respectively.

2) Parameter Analysis of Representative Band Selection: With respect to the representative band selection process, the parameter δ greatly affects the importance of the selected bands while performing the DL. To estimate the influence of the parameter δ, we fix the above two optimal parameters value m and λ on three datasets. As well, different parameters δ are set to study the impact on the performance of the proposed method when different number of bands are selected (the range of [5, 20]). Specifically, the parameter δ is set to 0.05, 0.1, 0.5, and 1, respectively.

The classification results about different parameters δ on three datasets are shown in Table IV. As can be seen in this table, for the Pavia University dataset, the accuracy is superior when $\delta = 0.5$, the maximum OA gap is more than 5%. With respect to the Botswana dataset, the classification accuracy is relatively good when the parameter δ is set to 0.05. Equally, the optimal parameter value δ is 0.05 on Salinas dataset.

D. Study of Block Diagonal Clustering

In our work, the block diagonal similarity matrix $\boldsymbol{S}$ is constructed by solving (15). To investigate the viability of this strategy and demonstrate the effort of block diagonal structure, the block diagonal similarity matrix is compared with the initial affinity matrix. Meanwhile, the matrix visualization results are provided on three hyperspectral datasets.

The experimental visualization results are shown in Figs. 3–5. They display the block diagonal clustering results of 5 and 10 clusters (i.e., subblock matrices) on Pavia University, Botswana, and Salinas datasets, respectively. It is clear that the obtained similarity matrix possesses a strictly block diagonal

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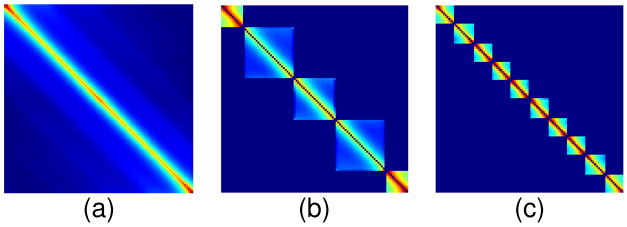


Fig. 3. Block diagonal clustering result on Pavia University dataset. (a) Initial affinity matrix. (b) and (c) Block diagonal similarity matrices with 5 and 10 subblocks, respectively.

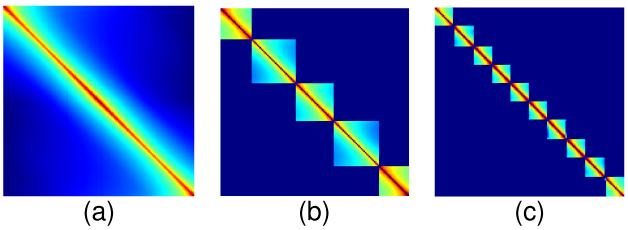


Fig. 4. Block diagonal clustering result on Botswana dataset. (a) Initial affinity matrix. (b) and (c) Block diagonal similarity matrices with 5 and 10 subblocks, respectively.

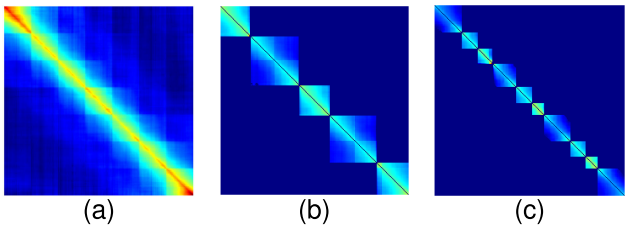


Fig. 5. Block diagonal clustering result on Salinas dataset. (a) Initial affinity matrix. (b) and (c) Block diagonal similarity matrices with 5 and 10 subblocks, respectively.

structure and data points are arranged in order simultaneously on different datasets. Therefore, each subblock matrix is available for subsequent representative band selection process. Overall, the proposed method can achieve the block diagonal clustering with the desired number of subblock matrices for hyperspectral band selection.

E. Ablation Study

With respect to the proposed method, it mainly consists of two parts: block diagonal partition (i.e., block diagonal generation) and representative band selection. To verify the efficiency of two parts, we conduct the experiment separately. According to the relevant literature, Wang et al. propose the FNGBS [50], which has a similar overall architecture to BDRLA. Therefore, FNGBS is utilized as the comparison algorithm in ablation study.

Specifically, after the block diagonal partition, the product of local density and information entropy for each band is calculated and ranked descendingly. The band with largest value in each cluster is selected to construct the band subset. For clarity, this algorithm is referred to BDRLA-FNGBS. Moreover, to illustrate the effectiveness of representative band selection, the FNGBS is adopted instead of our block diagonal partition method. Similarly, this algorithm is named FNGBS-BDRLA. Meanwhile, we set the band range of [5, 30] to estimate the performance of each part.

TABLE V
AOA OF TWO CLASSIFIERS ON THREE DATASETS FOR BLOCK DIAGONAL PARTITION (%)

Classifier	Method	Pavia University	Botswana	Salinas
SVM	BDRLA-FNGBS	86.83	83.46	89.10
	FNGBS [50]	81.06	84.53	87.38
*K*NN	BDRLA-FNGBS	85.77	80.78	87.78
	FNGBS [50]	83.90	80.20	86.90

TABLE VI
AOA OF TWO CLASSIFIERS ON THREE DATASETS FOR REPRESENTATIVE BAND SELECTION (%)

Classifier	Method	Pavia University	Botswana	Salinas
SVM	FNGBS-BDRLA	84.20	85.18	88.44
	FNGBS [50]	81.06	84.54	87.38
*K*NN	FNGBS-BDRLA	83.30	81.43	86.65
	FNGBS [50]	82.60	77.72	86.52

1) Effectiveness of Block Diagonal Partition: Fig. 6 shows the classification results for block diagonal partition part by SVM classifier on three datasets. Table V displays the AOA results of two classifiers on three datasets. As can be seen from the figure and table, the performance of FNGBS is worse than BDRLA-FNGBS. Especially in the case of selecting a few bands, the difference is more obvious. The reason is that FNGBS may ignore band selection in terms of global optimal. Overall, it demonstrates the superiority of this part in BDRLA further.

2) Effectiveness of Representative Band Selection: The OA curves for representative band selection part by KNN classifier are displayed in Fig. 7. The AOA results of two classifiers on three datasets are exhibited in Table VI. Evidently, the accuracy curve of FNGBS-BDRLA is higher than FNGBS with the increase of the number of bands on three datasets. Meanwhile, its accuracy exceeds the accuracy of all bands at certain locations on Pavia University and Botswana datasets. All in all, the representative band selection part has an excellent performance in this article.

F. Comparison Classification Performance

To thoroughly evaluate the advantage of BDRLA while comparing to five UBS algorithms (i.e., MVPCA, E-FDPC, FNGBS, VGBS, and OPBS), two classifiers are employed to analyze the HSI on three public datasets. The comparison contents include: 1) OA curves against different number of bands by two classifiers and 2) AOA bars about the range of 5–50 bands.

1) Pavia University: For this dataset, the classification results produced by the SVM and KNN classifiers are exhibited in Fig. 8. According to Fig. 8(a) and (b), BDRLA acquires the best classification performance compared with the other competitors when the number of bands ranges from 1 to 15. As the number of bands exceeds 15, BDRLA provides comparable classification performance with others. This result

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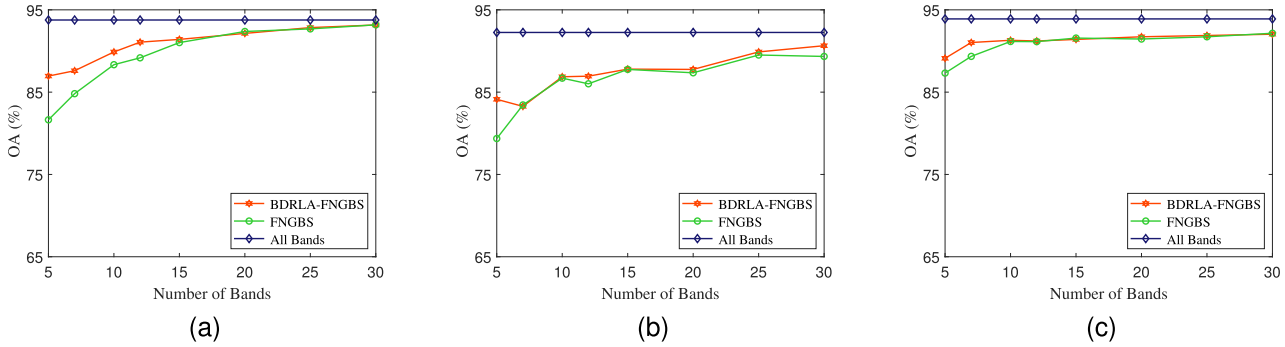


Fig. 6. Classification results for block diagonal partition part on three datasets. (a)–(c) Results by SVM classifier on Pavia University, Botswana, and Salinas datasets, respectively.

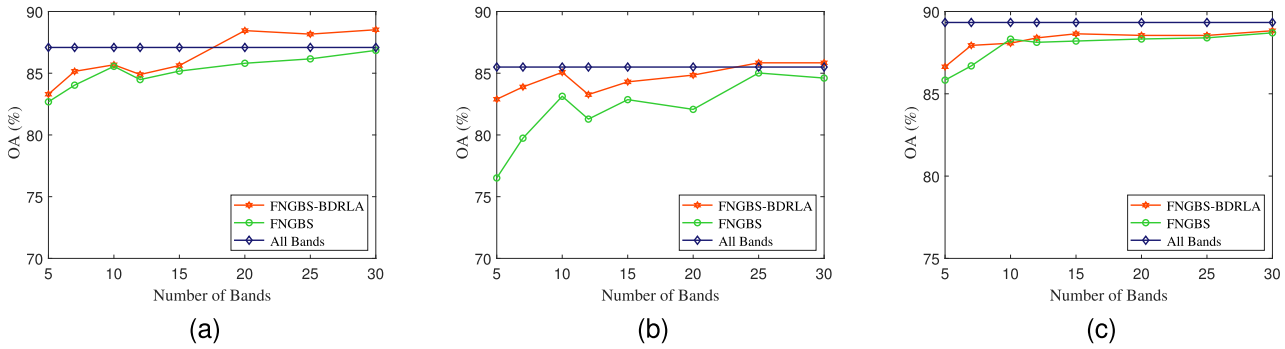


Fig. 7. Classification results for representative band selection part on three datasets. (a)–(c) Results by KNN classifier on Pavia University, Botswana, and Salinas datasets, respectively.

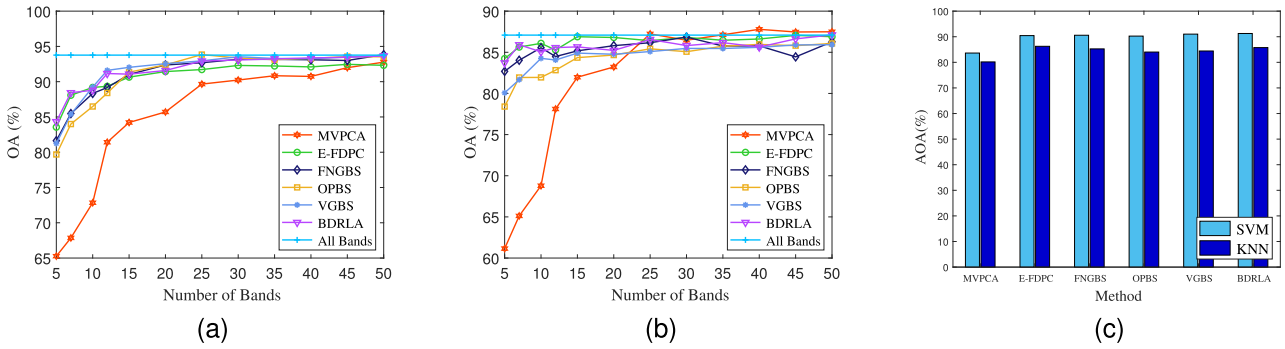


Fig. 8. Classification results by selecting different number of bands on Pavia University dataset. Results on (a) SVM and (b) KNN. (c) Average classification results.

does not cast doubt to the advantage of BDRLA, for the reason that the better performance of classification with fewer bands is more convincing in the hyperspectral band selection. Since E-FDPC does not consider the correlation between the adjacent bands, it obtains a lower AOA than BDRLA. FNGBS combines the clustering and ranking method. Therefore, it acquires better results than OPBS and VGBS. As for MVPCA, there is a substantial gap between it and other competitors. This verifies that ignoring the redundancy of selected bands results in an unsatisfactory performance. From Fig. 8(c), obviously, it can see that the classification result by BDRLA achieves the highest accuracy. In general, our method yields satisfactory classification results, which is comparable to the latest band selection algorithms.

2) Botswana: The experiment results of the proposed method about three aspects are shown in Fig. 9. Among all the methods, BDRLA provides superiority to the other methods in OA, and the performance is more significant when the number of selected bands is small. Specifically, when using the SVM classifier [see Fig. 9(a)], BDRLA obtains the highest accuracy, which can be competitive with the accuracy of using all bands in a certain location. Meanwhile, since the OPBS and VGBS all are improved based on the MEV method, they obtain approximately the same result. Moreover, MVPCA and FNGBS have great fluctuation in a small number of selected bands. This is due to the selection of adjacent bands with high correlation. On the other hand, when using the KNN classifier, BDRLA displays the best performances and is slightly superior to VGBS and OPBS in Fig. 9(b). Similar to Pavia University dataset, MVPCA performs not well. Fig. 9(c) shows the average classification accuracy of six algorithms, and BDRLA outperforms the best performance among the two

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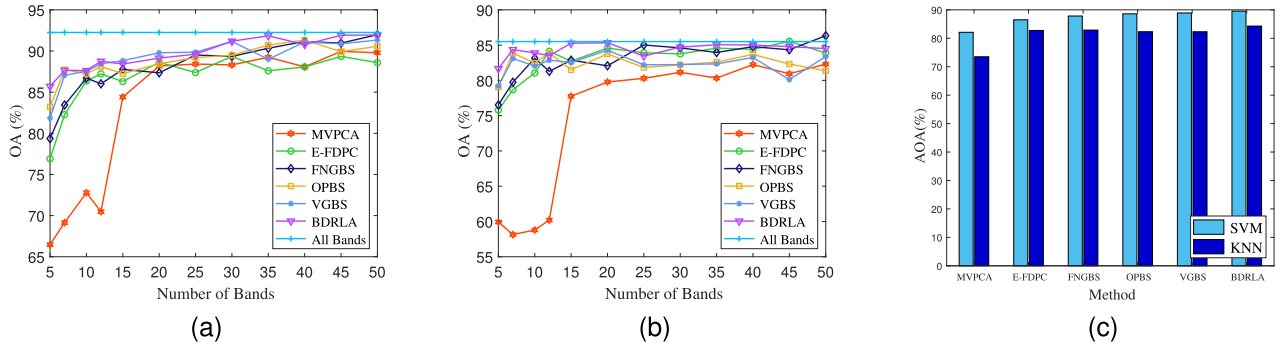


Fig. 9. Classification results by selecting different number of bands on Botswana dataset. Results on (a) SVM and (b) KNN. (c) Average classification results.

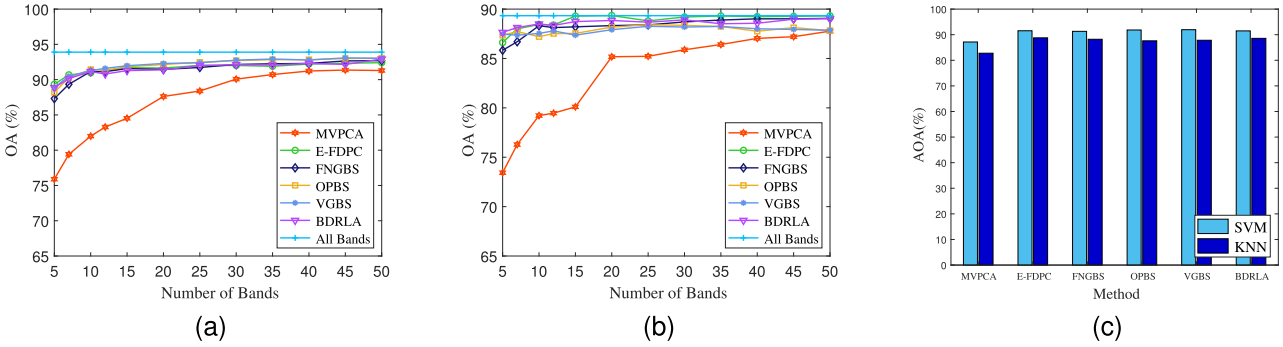


Fig. 10. Classification results by selecting different number of bands on Salinas dataset. Results on (a) SVM and (b) KNN. (c) Average classification results.

TABLE VII
PROCESSING TIME OF DIFFERENT METHODS ON THREE DATASETS (S)

Dataset	MVPCA [18]	E-FDPC [49]	FNGBS [50]	OPBS [51]	VGBS [52]	BDRLA
Pavia University	1.526	0.523	0.574	2.125	0.48	0.636
Botswana	2.625	1.176	1.19	4.378	1.134	1.347
Salinas	3.081	0.547	0.642	0.832	0.626	0.766

classifiers. On the whole, the experiment results further prove that the proposed method can achieve satisfactory performance for classification.

3) Salinas: As can be concluded in Fig. 10, the results of classification accuracy curves demonstrate that the classification accuracy increases as the number of the selected bands increases for all the band selection methods. Different from the above two datasets, all band selection algorithms obtain approximately the same OA except MVPCA, which is mainly due to the high correlation of the bands selected by the ranking-based methods. Under the SVM classifier [see Fig. 10(a)], BDRLA reveals a satisfactory performance and is superior to the other methods in general. When it comes to KNN classifier [see Fig. 10(b)], results are a little different. Specifically, when the number of selected bands goes beyond 10, BDRLA and FNGBS perform the best and they are slightly superior to OPBS, E-FDPC, and VGBS. Similar to Botswana dataset, BDRLA achieves the best performance when the number of selected bands is small. Additionally, when averaging the results, BDRLA behaves approximately the same result as E-FDPC and VGBS, but is accurate than the other approaches in Fig. 10(c). In summary, this experiment again verifies that the proposed method offers a significant advantage to other competitors.

By conducting abundant experiments on three HSI datasets, some key conclusion can be summarized. BDRLA attains better and stable performance across different datasets in general, and it indicates that the proposed method is robust enough for hyperspectral band selection task. Furthermore, BDRLA still achieves higher OA when a small number of bands is taken.

G. Computational Time Comparison

To verify the effectiveness of the proposed method further, the computational time of all competitors conducted on three datasets. For equal comparison, all algorithms perform in MATLAB R2021b with PC workstation (Intel Core i7-12700H CPU processor and 16-GB RAM). Table VII reveals the time taken by different methods to select five bands on different datasets. As observed, the computation time required by OPBS and MVPCA is larger than other competitors for three datasets. The reason for this is that they have to calculate the OP and covariance matrix separately. By analyzing the table in greater detail, one can discover that the difference of execution time of

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FNGBS, VGBS, and E-FDPC is between 0.1 and 0.2. It also can be seen that some algorithms (E-FDPC, OPBS, VGBS, etc.) take slightly smaller time on Pavia University dataset than on Botswana dataset. The reason is that the image size of the latter is larger than the former. Although our method is relatively time-consuming than E-FDPC and VGBS, it can achieve promising results in classification performance and other aspects. Please note that the execution time in this table is also approximately the maximum time. Additionally, the BDRLA algorithm needs to set a certain number of iterations in both block diagonal generation and band selection strategy to obtain the optimal solution. Therefore, the computation time is acceptable for the proposed method while satisfying better performance in classification accuracy.

V. Conclusion

In this article, a BDRLA is proposed to address the hyperspectral band selection. The fundamental idea of the proposed method is to obtain the high-quality similarity matrix and partition the several clusters with block diagonal structure simultaneously, which enhances the effectiveness of clustering than other graph clustering algorithms. For each cluster (i.e., subblock matrix), we handle the band vectors in the low-dimensional manifold space and perform representative band selection problem from a novel-ranking perspective. By doing so, excellent classification accuracy is obtained while reducing the redundant information of HSI. Intensive experiments illustrate that our algorithm achieves outstanding performance improvement compared with other competitors on three public datasets. In the future, we will extend the proposed method to construct the data similarity matrix directly and consider the optimization of graph matrix.

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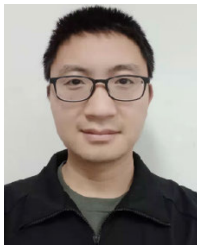
Shuying Li (Member, IEEE) received the Ph.D. degree in digital circuit and system from the Chinese Academy of Sciences, Shanghai, China, in 2010.

She is currently a Professor of control science and engineering with the School of Automation, Xi'an University of Posts and Telecommunications, Xi'an, China. Her research interests include remote sensing, computer vision, and pattern recognition.



Zhe Liu received the B.E. degree in automation from the Xi'an University of Posts and Telecommunications, Xi'an, China, in 2021, where he is currently pursuing the M.S. degree with the School of Automation.

His research interests include hyperspectral image processing and pattern recognition.



Long Fang received the Ph.D. degree in engineering from Northwestern Polytechnical University, Xi'an, China, in 2020.

He is currently an Research Assistant with the Northwest Institute of Nuclear Technology, Xi'an. His research interests include computer vision and remote sensing.



Qiang Li (Member, IEEE) received the Ph.D. degree in computer science and technology from Northwestern Polytechnical University, Xi'an, China, in 2022.

He is currently a Post-Doctoral Researcher with the School of Electronic Engineering, Xidian University, Xi'an. His research interests include remote sensing image processing and computer vision.

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